Shah Md Nehal Hasnaeen

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Research and Teaching Experience

Graduate Research Assistant, University of Tennessee Knoxville Aug 2023 – Feb 2026

Investigated socially assistive robotic systems for Alzheimer's and dementia care, progressing from teleoperated platforms to near-autonomous AI pipelines, with parallel work in multimodal computer vision.

- ***FujiView: Multimodal Late-Fusion for Scenic Visibility Prediction*** 2025–2026
  Published — IEEE/CVF WACV 2026
  Multimodal late-fusion of vision and weather forecast features for scenic visibility prediction; ACC ≈ 0.892, AUC ≈ 0.94 across a 113,000-image, 42-camera dataset.
- ***RL Simulation Framework for Adaptive Dementia Care*** 2025
  Published — arXiv:2501.17206; Poster, IEEE BHI 2025
  Two-agent RL simulation (PLWD + robot caregiver) for adaptive dementia care
- ***Generative AI in Healthcare: Ethical and Practical Challenges*** 2025
  Published — Frontiers in Digital Health, vol. 7 (credited for Supervision)
  Supervised the first author of a PRISMA systematic review (54 studies) from design through publication; directed the paper's structure and analytical framework
- ***Portable PASS: Teleoperation-Based SAR Platform for ADL Assistance*** 2026
  Under Review — ACM Transactions on Computer-Human Interaction (TOCHI)
  Wizard-of-Oz SAR platform (Pepper robot, 8 ADL tasks); IRB-approved usability study with 10 older adults; produced an annotated multimodal dataset for downstream model training.

Graduate Research Assistant, Idaho State University May 2022 – Aug 2023

Investigated physical-layer security for UHF RFID systems, developing electromagnetic fingerprinting-based machine learning methods for malware detection and digital twin identification in IoT infrastructure.

- ***UHF RFID EM Fingerprint-Based Digital Twin Web Resolver*** 2022–2023
  Published — IEEE RFID 2023 (first author); Poster, IEEE DTPI 2022 (HeroX Honorable Mention)
  Proposed EM fingerprinting as an unclonable identifier for resolving physical RFID tags to cloud-hosted digital twins; 95–99% classification accuracy across tag types.
- ***ML for Longitudinally Slotted Waveguide Characterisation*** 2023
  Published — IEEE APS/URSI 2023 (co-authored)
  Co-developed an ML-based characterisation model for a longitudinally slotted waveguide embedded in a high-dielectric medium.
- ***RFID Malware Detection via Physical-Layer EM Fingerprinting*** 2021–2022
  Published — IEEE RFID 2022 (first author)
  Physical-layer SQL-injection malware detection from RFID EM signatures without memory access; proposed two novel spectral feature-reduction methods; 82.5% test accuracy.

Graduate Teaching Assistant, Idaho State University Aug 2021 – May 2022

- *Courses:* ECE 2200/3300 (Electric Circuits I/II), ECE 2250 (Digital Systems), and ECE 3340 (Electromagnetics), including associated laboratory sections.
- *Responsibilities:* Held regular office hours, provided one-on-one tutoring, and graded assignments and exams across three semesters under multiple instructors.

Miscellaneous Projects

Load Scheduling of a Refrigerated Warehouse with Homogeneous Compressors 2019
Published — ICASERT 2019 (first author; bachelor's thesis, BUET EEE)
Proposed uniform-delay compressor scheduling for refrigerated warehouses under dynamic pricing; derived analytical load and cost models validated to $<1\%$ error; demonstrated 27% electricity cost reduction vs. random-start operation.

Education

Idaho State University, M.S., Electrical and Computer Engineering 2021 – 2023
Focus: RFID Electromagnetic Fingerprinting, SQL Injection Malware Detection, Digital Twin Identification
CGPA: 3.96 out of 4.00

Bangladesh University of Engineering and Technology, B.S., Electrical and Electronic Engineering 2015 – 2019
Major: Communication, Minor: Electronics

SKILLS

Programming & Frameworks: Python, MATLAB, R, C/C++; PyTorch, HuggingFace Transformers, OpenCV, scikit-learn

Computer Vision & Multimodal Learning: Object detection, video understanding, vision-language models (VLMs), model fine-tuning and adaptation, action recognition, multimodal late-fusion, task graph construction

Deep Reinforcement Learning: Q-learning, policy gradient methods, reward shaping, policy evaluation, multi-agent simulation environment design

Robotics & Human-Robot Interaction: Humanoid robot programming, Wizard-of-Oz teleoperation, IRB protocol design, multimodal data collection (video, audio, physiological signals)

RF Engineering & Signal Processing: UHF RFID systems, spectrum analysis, electromagnetic fingerprinting, frequency-domain feature extraction

Machine Learning: Supervised and unsupervised learning, feature engineering, dimensionality reduction

High-Performance Computing: GPU cluster environments, distributed model training on HPC systems

Data Science & Research Methods: Experimental design, quantitative and qualitative analysis, dataset annotation and pipeline development, benchmark construction

LIST OF PUBLICATIONS

1. B. Bible, S. M. N. Hasnaeen, and H. Qi, "FujiView: Multimodal Late-Fusion for Predicting Scenic Visibility," in *Proc. IEEE/CVF WACV*, 2026. (Accepted)
2. T. Tung, S. M. N. Hasnaeen, and X. Zhao, "Ethical and Practical Challenges of Generative AI in Healthcare and Proposed Solutions: A Survey," *Frontiers in Digital Health*, vol. 7, 2025.
3. S. M. N. Hasnaeen, S. Neupane, and A. Chrysler, "Towards a UHF RFID Electromagnetic Fingerprint based Web Resolver for Digital Twins," in *Proc. IEEE RFID*, 2023, pp. 36–41.
4. Z. Mohammad, S. M. N. Hasnaeen, S. Neupane, and A. Chrysler, "Machine Learning Approach to Determine Characteristics of a Longitudinally Slotted Waveguide in High Dielectric," in *Proc. IEEE APS/URSI*, 2023.
5. S. M. N. Hasnaeen and A. Chrysler, "Detection of Malware in UHF RFID User Memory Bank using Random Forest Classifier," in *Proc. IEEE RFID*, 2022, pp. 47–52.
6. S. M. N. Hasnaeen, S. Rahman, and M. F. Uddin, "Load Scheduling of a Refrigerated Warehouse with Homogeneous Compressors Under Dynamic Pricing," in *Proc. ICASERT*, 2019.
7. F. Yuan, S. M. N. Hasnaeen, T. Morris, B. Bible, R. P. Lopez, M. Oliver, J. G. Anderson, M. Crane, A. Graff, D. Horn, H. Qi, and X. Zhao, "A Field-Deployable Human-Robot Interaction Platform to Assist Older Adults During Complex Activities of Daily Living," Submitted to *ACM Transactions on Computer-Human Interaction Journal* for review.
8. F. Yuan, S. M. N. Hasnaeen, R. Zhang, B. Bible, J. R. Taylor, H. Qi, F. Yao, and X. Zhao, "Integrating Reinforcement Learning and AI Agents for Adaptive Robotic Interaction and Assistance in Dementia Care," *arXiv preprint arXiv:2501.17206*, 2025.

PRESENTED POSTERS

1. F. Yuan, S. M. N. Hasnaeen, R. Zhang, B. Bible, J. R. Taylor, H. Qi, F. Yao, and X. Zhao, "Integrating Reinforcement Learning and AI Agents for Adaptive Robotic Interaction and Assistance in Dementia Care," *IEEE BHI*, 2025.
2. S. M. N. Hasnaeen, S. Neupane, and A. Chrysler, "UHF RFID Electromagnetic Fingerprint-Based Digital Twin Resolver," *IEEE DTPI*, 2022.

AWARDS

Best Graduate Researcher, Department of Electrical and Computer Engineering, 2023
Awarded by College of Science and Engineering, Idaho State Univesity

HeroX Digital Twin Resolver Challenge – Honorable Mention, IEEE DTPI Conference, 2022
Sole student team award for "UHF RFID Electromagnetic Fingerprint-Based Digital Twin Resolver"

REFERENCES

Andrew Chrysler
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Los Alamos, NM 87545
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Fengpei (Fiona) Yuan
Assistant Professor
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Worcester Polytechnic Institute, Worcester, MA 01609
Relationship: Collaborator
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